Hasan Zakeri

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SUMMARY

Machine Learning Researcher with 8+ years of specialized experience developing AI for real-time autonomous systems and safety-critical control. Deep expertise in advanced deep learning techniques, including Reinforcement Learning (RL), imitation learning, and Bayesian Neural Networks for uncertainty quantificationfor complex decision-making. Skilled at building and optimizing scalable ML pipelines in a diverse range of real world problems. Five years of industry experience as a full-stack developer providing a strong foundation for deploying high-performance software architectures.

SKILLS AND CERTIFICATES

Machine Learning: Supervised/unsupervised learning, reinforcement learning, deep learning, statistical analysis, LLM fine-tuning, RAG, prompt engineering, transformers.
Programming: Python (PyTorch, TensorFlow, Scikit-Learn, Keras, NumPy, Pandas), C++, SQL, React, Django.
Data Systems: Spark, Hadoop, ETL pipelines, large-scale data processing.
Tools & Platforms: AWS, CI/CD pipelines, MySQL, Embedded Systems.
Certificates: Designing Microsoft Azure Infrastructure Solutions (AZ-305), Implementing Analytics Solutions Using Microsoft Fabric (DP-600), Generative AI with Large Language Models, Introduction to Gen-AI, Gen-AI with Large Language Models, Build a Backend REST API with Python & Django, Python Django - The Full Stack Guide, Amateur Radio Extra Class
Simulation Tools: Matlab/Simulink, CRALA, pyChrono, PyBullet, Airsim, CarSim
Soft Skills: Research, cross-functional collaboration, leadership.

EDUCATION

Illinois Institute of Technology — Chicago, IL
Ph.D. in Mechanical and Aerospace Engineering — (Expected) Jan 2026
University of Notre Dame — Notre Dame, IN
M.S. in Electrical Engineering — Advisor: Panos Antsaklis

EXPERIENCE

Illinois Institute of Technology, *Graduate Research Assistant* — Chicago, IL, Ongoing
- Developed a deep learning framework achieving 35% improved uncertainty quantification for autonomous vehicles.
- Built multi-agent reinforcement learning framework that improves traffic flow by 40% and reduces congestion time by 30%.

University of Notre Dame, *Graduate Research Assistant* — Notre Dame, IN, 2016–2019
- Designed data driven resilient controller achieving 95% attack detection accuracy using unsupervised machine learning.
- Developed polynomial approximation framework reducing computational complexity by 50% while maintaining performance.

EmNet LLC., *Machine Learning Research Intern* — South Bend, IN, 2016
- Created deep learning architecture improving sewer water management prediction accuracy by 40% across 4 major US cities.
- Implemented neural network control mechanism in C for EPA-SWMM, deployed in production systems across multiple cities.

Tachara Online Bookstore, *Technical Project Lead* — Tehran, Iran, 2010–2013
- Designed an e-commerce big data processing platform using .NET MVC 4, C# and SQL Server serving over 1M active users.
- Integrated payment gateways and inventory systems reducing transaction processing time by 60%.

Azar Negar Shargh Co., *Principal Engineer* — Shiraz, Iran, 2004–2009
- Architected and developed responsive, full-stack content management systems for high-profile corporate communications.
- Led cross-functional engineering teams implementing agile methodologies enabling same-day feature deployment.

MACHINE LEARNING AND CONTROL PROJECTS

Traffic Congestion using Deep RL | PyTorch, PPO, Multi-Head Attention, Actor-Critic Networks, GNN — Ongoing
- Attention-based PPO agent to optimize highway traffic flow through speed commands, achieving 25–40% congestion reduction
- Designed using multi-head attention state to handle 10–100+ vehicles with lane-change and graph-based highway topology

Autonomous Vehicles Control with RL | *CARLA simulator, Stable Baselines, PyTorch, OpenCV* — Spring 2023
- Trained a reinforcement learning agent using CARLA simulator images to autonomously control vehicle navigation.
- Developed a vision-based controller that learned driving in a realistic urban environment without explicit programming.

Vehicle Dynamics Model: Uncertainty Quantification | pyChrono, scikit, PyTorch, torchbnn — Winter 2022
- Validated first-principles vehicle dynamics models against Chrono::Vehicle simulations and quantified modeling uncertainty.
- Characterized modeling uncertainties using Gaussian Process Regression and Bayseian neural with 80% improvement.

Distributed Filter Design Agent | *Vision Transformers, YOLO, ResNet, SFT, ADS, HFSS* — Winter 2024
- Trained multi-modal Vision Transformer with YOLO and ResNet on filter topologies with 92% accuracy after post-training.
- Developed end-to-end AI agent for synthesizing novel filters from desired characteristics, reducing the iteration time by 85%.

RF Signal Classification Assistant | *PyTorch, Transformers, Signal Processing* — Spring 2024
- Developed multimodal LLM combined with spectral analysis to classify and identify 95% of unknown RF signals.

- Built a conversational interface to query signal characteristics using natural language, reducing the analysis time by 50% %.

Data-driven Controller Reconfiguration for Fault Mitigation

- A data-driven adaptive controller reconfiguration that detects and mitigates malfunctions using only input and output data.
- Ensures closed-loop stability and fault-tolerance in autonomous systems by a real-time interface around the existing controller

SELECTED PUBLICATIONS

- **Hasan Zakeri** and Baisravan HomChaudhuri, "Graph Neural Network and Attention-Based Deep Reinforcement Learning for Heterogeneous Traffic Congestion Control with Variable Fleet Dynamics," In preparation for The International Conference on Machine Learning (ICML 2026).
- **Hasan Zakeri** and Baisravan HomChaudhuri, "Learning-Based Predictive Control Method for Vehicle Lateral Control with a Multi-Step Gaussian Regression Prediction," In preparation for The Conference on Robot Learning (CoRL 2026).
- **Hasan Zakeri** and Baisravan HomChaudhuri, "Learning-Based Predictive Control Method for Vehicle Lateral Control with a Multi-Step Gaussian Regression Prediction," submitted, Journal of Dynamic Systems, Measurement, and Control, 2025.
- **Hasan Zakeri** and Baisravan HomChaudhuri, "Iterative Stochastic Model Predictive Control with Multi-Step Uncertainty Learning Model: Vehicle Lane Change Case Study," Modeling, Estimation and Control Conference, MECC 2024.
- **Hasan Zakeri** and Baisravan HomChaudhuri, "Multi-Step Gaussian Regression Prediction in Dynamic Systems: A Case Study in Vehicle Lateral Control," in *ASME Letters in Dynamic Systems and Control* vol. 3, no. 4, Oct. 2023.
- **Hasan Zakeri** and Panos J. Antsaklis, "Passivity Measures in Cyberphysical Systems Design: An Overview of Recent Results and Applications," in *IEEE Control Systems Magazine,* vol. 42, no. 2, pp. 118-130, April 2022.
- **Hasan Zakeri** and Panos J. Antsaklis, "Passivity and Passivity Indices for Switched and Cyber-Physical Systems," (invited) in Encyclopedia of Systems and Control, J. Baillieul and T. Samad, Eds. London: Springer London, 2020.
- **Hasan Zakeri** and Panos J. Antsaklis, "Passivity, Dissipativity and Passivity Indices," (invited) in Encyclopedia of Systems and Control, J. Baillieul and T. Samad, Eds. London: Springer London, 2020.
- **Hasan Zakeri** and Panos J. Antsaklis, Passivity and Passivity Indices of Nonlinear Systems under Operational Limitations using Approximations, International Journal of Control, July 2019.
- **Hasan Zakeri** and Panos J. Antsaklis, A Data-driven Adaptive Controller Reconfiguration for Fault Mitigation: A Passivity Approach, 27th Mediterranean Conference on Control and Automation (MED), Akko, Israel, 2019, pp. 25–30.
- More at `https://scholar.google.com/citations?user=a-Z05tMAAAAJ`

LEADERSHIP AND SERVICES

- Member of search committee for Graduate Career Director position, reviewed and interviewed 6 candidates. University of Notre Dame — January 2018
- Professional Development Chair, Graduate Student Union, University of Notre Dame — 2017–2018
- Member of University Committee on Internationalization, appointed by the President, University of Notre Dame. — 2017–2018
- Jury and Test Designer for Iran's National Skills Olympiad on IT Software Solutions for Business — 2013